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### Task #1

A baseball is thrown by a pitcher with a velocity of 43 m/s. The batter hits **straight back** at the pitcher with a velocity of 56 m/s. If the ball was in contact with the bat for 0.0057 seconds and has a mass of 0.145 kg, what is the force on the baseball?

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### Task #2

While washing your car, water comes out of the hose at a rate of 1.5 kg/s and a speed of 20 m/s. How much force does the water impart to your car?

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### Task #3

Rain is falling at a rate of 2.5 cm/hr and accumulates in a pan. If the raindrops hit at 8.0 m/s, estimate the force on the bottom of the 1.0-square-meter pan due to the impacting rain which we assume does not rebound. The density of water is 1000 kg/m<sup>3</sup>.

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